CS-401

Parallel and Distributed Computing

Lecture 2

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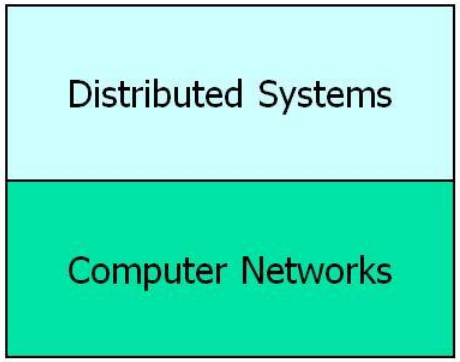
Defining Distributed Systems

- *"A system in which hardware or software components located at networked computers communicate and coordinate their actions only by message passing."* [Coulouris]
- *"A distributed system is a collection of independent computers that appear to the users of the system as a single computer."* [Tanenbaum]
- Example Distributed Systems:
  - Cluster:
    - *"A type of parallel or distributed processing system, which consists of a collection of interconnected stand-alone computers cooperatively working together as a single, integrated computing resource" [Buyya].*
  - Cloud: Virtualization pool of resources
    - *"a type of parallel and distributed system consisting of a collection of interconnected and **virtualised computers** that are dynamically provisioned and presented as one or more unified computing resources based on service-level agreements established through negotiation between the service provider and consumers" [Buyya].*

- Grid: A collection of computer resources from multiple locations to reach a common goal.

Networks vs. Distributed Systems

- Networks: A media for interconnecting local and wide area computers and exchange messages based on protocols. Network entities are visible and they are explicitly addressed (IP address).
- Distributed System: existence of multiple autonomous computers is transparent
- However,
  - many problems (e.g., openness, reliability) in common, but at different levels.
    - Networks focuses on packets, routing, etc., whereas distributed systems focus on applications.
    - Every distributed system relies on services provided by a computer network.



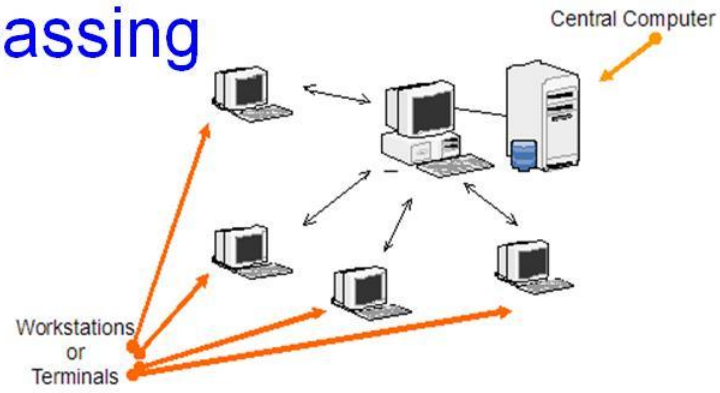
Cluster, Grid and Cloud computing

- Cluster: a group of computers are linked together so that they can act as a single entity.
- Grid: collection of computer resources from multiple locations to reach a common goal. Not a specific owner.
  - Resources are used in collaborative pattern. Less flexible than cloud. Low accessible service than cloud, application oriented
  - specialized software on every computer in the grid, software assigns subtasks to each computer
  - Example: Online gaming,
- Cloud: sharing computer resources rather than having local servers or personal devices. Owned by a single owner.
  - Is a pay-use business. More flexible, high accessible service, service oriented
  - Examples: Amazon Web Services (AWS), Google App Engine.
  - Example of cloud computing services: Dropbox, Gmail, Facebook, Youtube, RapidShare

- Cluster differs from Cloud and Grid in that a cluster is a group of computers connected by a local area network (LAN), whereas cloud and grid are more wide scale and can be geographically distributed. Another way to put it is to say that a cluster is tightly coupled, whereas a Grid or a cloud is loosely coupled. Also, clusters are made up of machines with similar hardware, whereas clouds and grids are made up of machines with possibly very different hardware configurations.

Characteristics of Distributed Systems

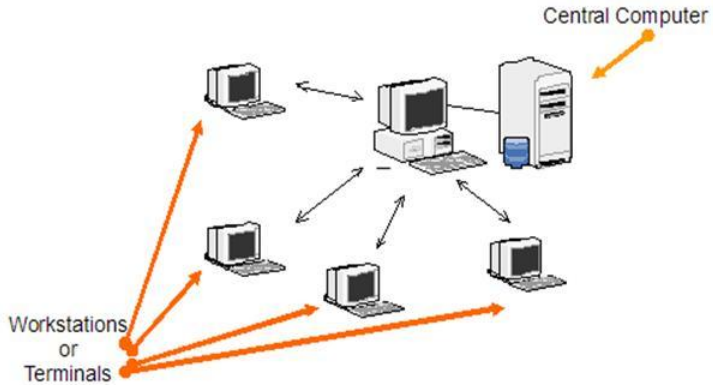
- Parallel activities
  - Autonomous components executing concurrent tasks
- Communication via message passing
  - No shared memory
- Resource sharing
  - Printer, database, other services
- No global clock
  - Only limited precision for processes to synchronize their clocks

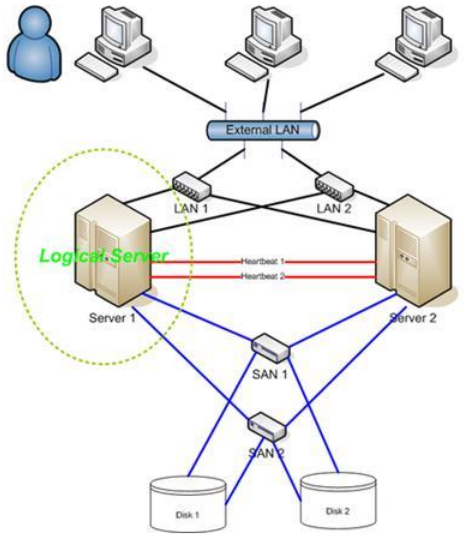


Goals of Distributed Systems

- Connecting Users and Resources
- Transparency
- Openness
- Scalability
- Enhanced Availability







Examples of Distributed Systems

- They (DS) are based on familiar and widely used computer networks:
  - Internet
  - Intranets, and
  - Wireless networks



- Example DS and its Applications:
  - Web (and many of its applications like Online bookshop)
  - Data Centers and Clouds
  - Wide area storage systems
  - Banking Systems
  - User-level communication (Facebook)





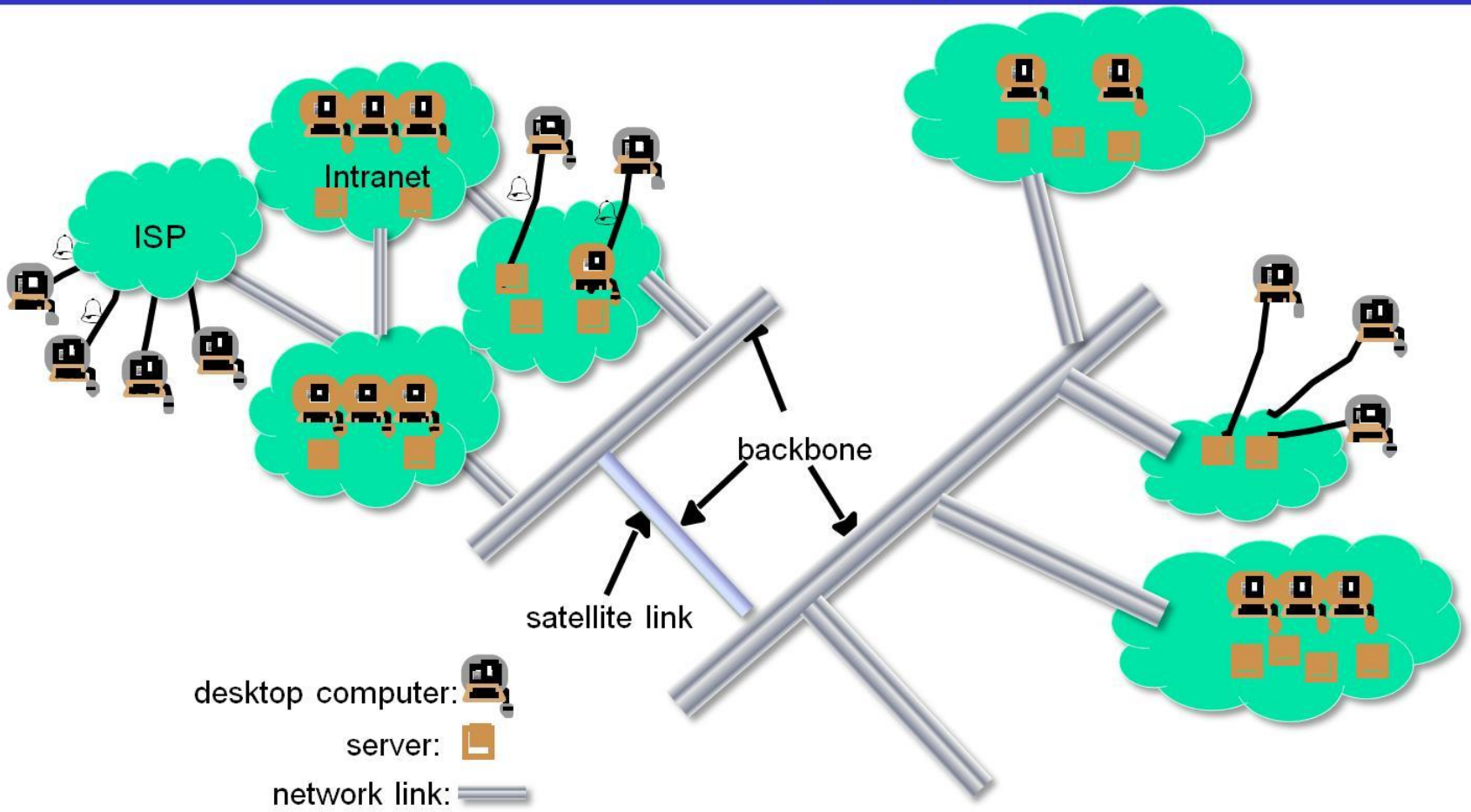
Selected application domains and associated networked applications

Finance and Commerce	eCommerce e.g. Amazon and eBay, PayPal, online banking and trading
The information Society	Web information and search engines, ebooks, Wikipedia; social networking: Facebook and Twitter.
Creative Industries and Entertainment	Online gaming, music and film in the home, user-generated content, e.g. YouTube, Flickr
Healthcare	Health informatics, on online patient records, monitoring patients (Metro South Health hospital trial in Queensland)
Education	e-learning, virtual learning environments; distance learning. e.g., Coursera
Transport and Logistics	GPS in route finding systems, map services: Google Maps, Google Earth
Science and Engineering	Cloud computing as an enabling technology for collaboration between scientists (LHC, LIGO)
Environmental Management	Sensor networks to monitor earthquakes, floods or tsunamis (Bureau of Meteorology flood warning system)

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A typical portion of the Internet and its services:

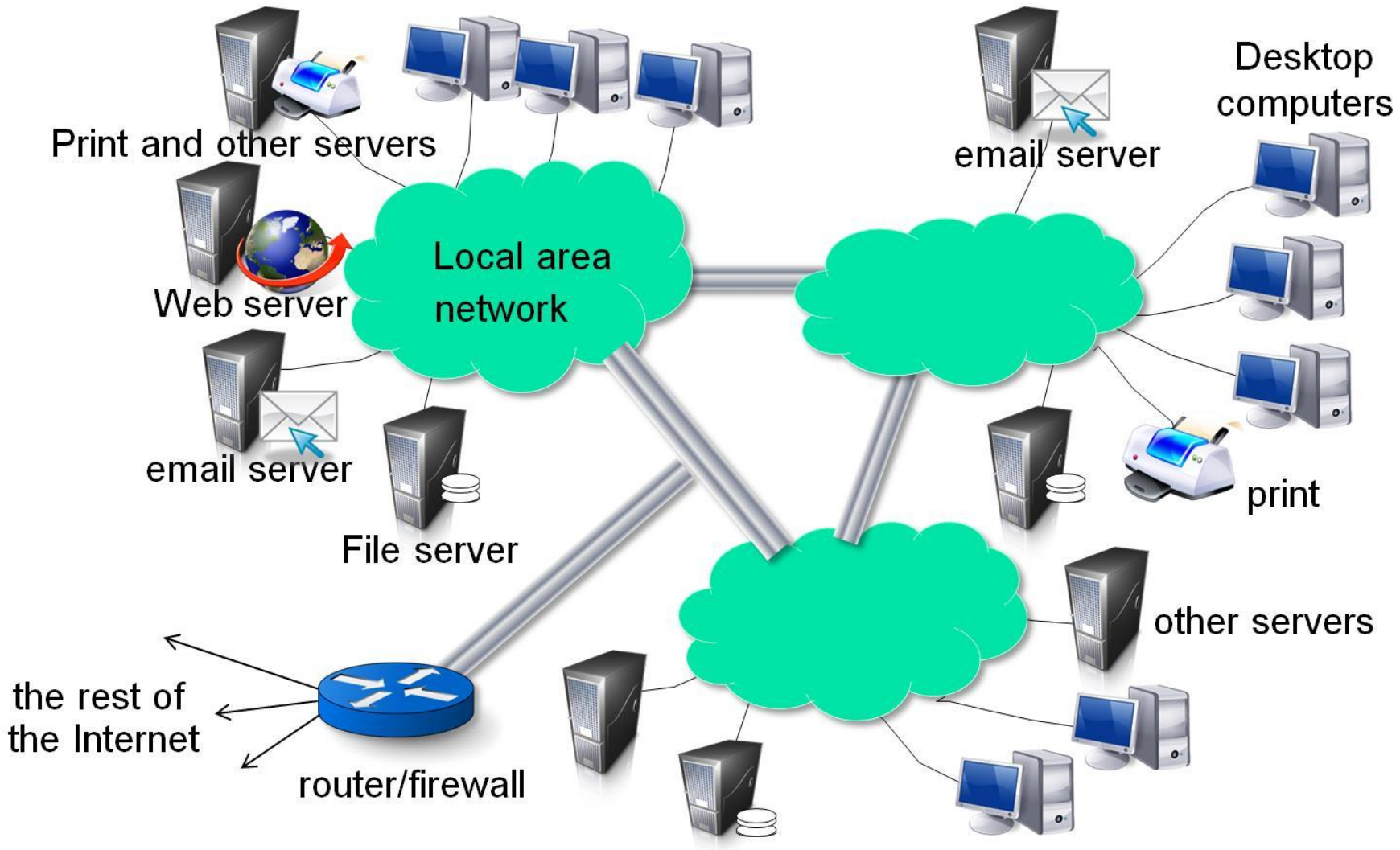
Multimedia services providing access to music, radio, TV channels, and video conferencing supporting several users.



- The Internet is a vast collection of computer networks of many different types and hosts various types of services.

A typical Intranet:

A portion of Internet that is separately administered & supports internal sharing of resources (file/storage systems and printers)



Overview Challenges I

- Heterogeneity
  - Heterogeneous components must be able to interoperate
- Distribution transparency
  - Distribution should be hidden from the user as much as possible
- Fault tolerance
  - Failure of a component (partial failure) should not result in failure of the whole system
- Scalability
  - System should work efficiently with an increasing number of users
  - System performance should increase with inclusion of additional resources

Overview Challenges II

- Concurrency
  - Shared access to resources must be possible
- Openness
  - Interfaces should be publicly available to ease inclusion of new components
- Security
  - The system should only be used in the way intended

Heterogeneity

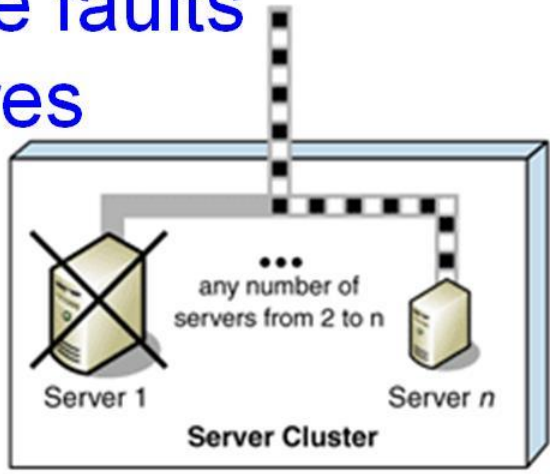
- Heterogeneous components must be able to interoperate across different:
  - Operating systems
  - Hardware architectures
  - Communication architectures
  - Programming languages
  - Software interfaces
  - Security measures
  - Information representation





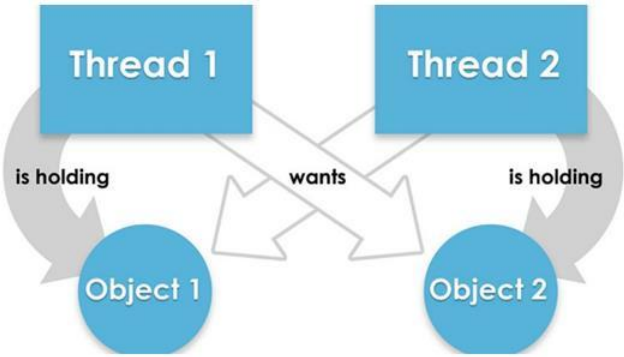
Fault Tolerance

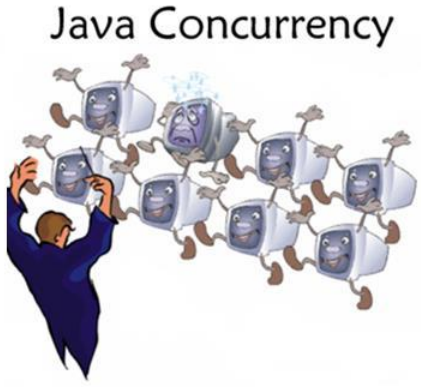
- Failure: an offered service no longer complies with its specification (e.g., no longer available or very slow to be usable)
- Fault: cause of a failure (e.g. crash of a component)
- Fault tolerance: no failure despite faults i.e., programmed to handle failures and hides them from users.



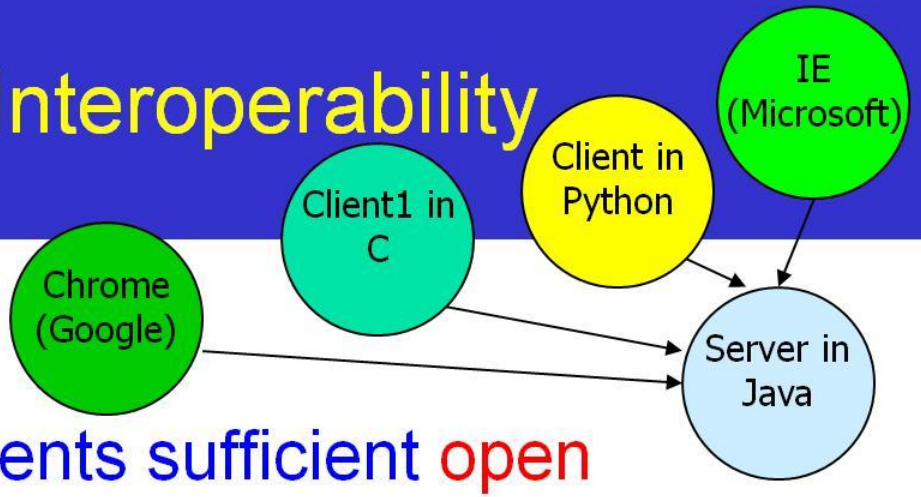
Concurrency

- Provide and manage concurrent access to shared resources:
  - Fair scheduling
  - Preserve dependencies (e.g. distributed transactions -- buy a book using Credit card, make sure user has sufficient funds prior to finalizing order)
  - Avoid deadlocks





Openness and Interoperability



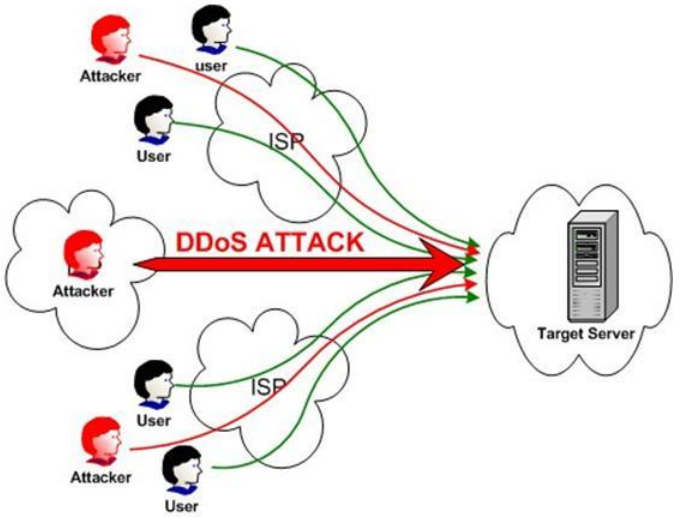
- Open system:
  "... a system that implements sufficient open specifications for interfaces, services, and supporting formats to enable properly engineered applications software to be ported across a wide range of systems with minimal changes, to interoperate with other applications on local and remote systems, and to interact with users in a style which facilitates user portability" (POSIX Open Systems Environment, IEEE POSIX 1003.0)
- Open spec/standard developers - communities:
  - ANSI, IETF, W3C, ISO, IEEE, OMG, Trade associations,...

Security I

- Resources are accessible to authorized users and used in the way they are intended
- Confidentiality
  - Protection against disclosure to unauthorized individual information
  - E.g. ACLs (access control lists) to provide authorized access to information
- Integrity
  - Protection against alteration or corruption
  - E.g. changing the account number or amount value in a money order

Security II

- Availability
  - Protection against interference targeting access to the resources.
  - E.g. denial of service (DoS, DDoS) attacks
- Non-repudiation
  - Proof of sending / receiving an information
  - E.g. digital signature



Security Mechanisms

- Encryption
  - E.g. Blowfish, RSA
- Authentication
  - E.g. password, public key authentication
- Authorization
  - E.g. access control lists

